Q 29.

- Sensor calibration is the process of adjusting sensor output to match the actual or known standard values.
  It is important in IoT systems because it ensures accurate and reliable data, which leads to better decision-making, automation, and efficiency.

- Calibrating an analog soil moisture sensor to get accurate percentage readings:
  - The sensor gives analog values (0-1023).
  - We need to map these values to 0% (dry) to 100% (wet).

- Two-point calibration method:

→ Step 1 : Dry calibration (0%).
  - Place the sensor in completely dry soil.
  - Note the analog value (Dry Value).
  - This represents 0% moisture.

→ Step 2 : Wet calibration (100%)
  - Place the sensor in fully saturated (wet) soil or water.
  - Note the analog value (Wet Value).
  - This represents 100% moisture.

→ Step 3 : Mapping the values
  - Use the formula to convert any analog reading to percentage:

$$\text{Moisture } (\%) = \left(\frac{\text{Analog Value} - \text{Dry Value}}{\text{Wet Value} - \text{Dry Value}} \right) \times 100$$

  - Limit the result between 0% and 100%.

→ Now the sensor will give accurate percentage moisture readings.